

Spatio-Temporal Analysis of Air Quality Disparities Across Urban and Rural India

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Abstract—Air pollution is usually treated as an urban problem, and rural regions are presumed to be cleaner. This analysis examines air quality in five pairs of urban-rural locations paired according to their proximity in India and calculates AQI based on the *Central Pollution Control Board (CPCB) Air Quality Index (AQI)* framework. Data from *September 2022 to January 2025* were utilized to evaluate key pollutants ($PM_{2.5}$, PM_{10} , CO , NO_2 , SO_2 , O_3), with daily AQI calculated from hourly measurements. *Spatio-temporal decomposition, Random Forest Regression, and K-Means Clustering* were utilized to find pollution drivers and trends. The results indicate that $PM_{2.5}$ was high in winter in cities while ozone levels were high in summer in rural areas. The *pollution disparity index (PDI)* calculates such disparities, confirming the necessity for *targeted policies, satellite-based monitoring and predictive modeling* to tackle the problem successfully.

Index Terms—Air Quality Index (AQI), Spatio-temporal analysis, Random Forest, K-Means clustering, Pollution Disparity Index (PDI), Seasonal air pollution trends, Air pollution monitoring, CPCB standards, Feature importance, Urban-rural air quality

I. INTRODUCTION

Air pollution continues to be an important environmental and public health problem, especially in fast-growing cities of India, where industrial growth, vehicle exhausts, and agricultural practices exacerbate air degradation. Although city centers are heavily affected by pollution caused by high population density and localized anthropogenic activities, rural areas are traditionally thought to enjoy cleaner air. But this assumption ignores pollution spillover from neighboring cities, biomass burning, dust storms, and other local sources that play a major role in rural air quality. It is important to understand these spatial and temporal variations in order to design effective region-specific mitigation strategies.

To scientifically evaluate these differences, this study investigates trends in air quality in five sets of urban-rural location pairs in India based on standardized Air Quality Index (AQI) calculations following the guidelines of the *Central Pollution Control Board (CPCB)*. AQI has been calculated with the sub-index approach where the concentrations of the

pollutants ($PM_{2.5}$, PM_{10} , CO , NO_2 , SO_2 , O_3) are translated into sub-indices following CPCB-established breakpoints and the maximum sub-index defines the final AQI. This method offers a uniform, comparable measure of pollution intensity between sites.

The dataset, spanning *September 2022 to January 2025*, is sourced from the *Open-Meteo API*, which provides high-resolution air quality and meteorological data based on geo-coordinates. Hourly pollutant concentrations are aggregated into daily AQI values, ensuring alignment with CPCB reporting standards. Additionally, geospatial factors such as proximity to urban centers and industrial zones are considered to analyze pollution variability.

To accurately capture spatio-temporal pollution dynamics, the research combines several analytical techniques. Descriptive and comparative analysis emphasize pollution trends, while spatio-temporal decomposition methods determine long-term trends and episodic pollution peaks. *Random Forest Regression* identifies major drivers of AQI variation, providing information on pollutant contributions over time. Furthermore, *K-Means clustering* categorizes areas according to AQI severity, demonstrating clear urban-rural pollution patterns. In order to measure the disparity in urban-rural pollution, the work formulates the *Pollution Disparity Index (PDI)*, presenting an objective indication of pollution spillover across densely located urban and rural areas. These methodologies facilitate a systematic estimation of the variation in pollution between various environments and seasons.

This research presents critical evidence to policymakers and environmental authorities, stressing the necessity of *location-specific air quality management policies* over blanket policies that do not consider localized pollution sources and transport mechanisms. In addition, the research highlights the necessity of *seasonally adaptive pollution control policies*, as urban $PM_{2.5}$ concentrations are highest during winter, whereas rural ozone concentrations increase during summer.

TABLE I: Pollutant Abbreviations and Their Full Forms

Abbreviation	Full Form
PM _{2.5}	Particulate Matter (diameter $\leq 2.5\mu\text{m}$)
PM ₁₀	Particulate Matter (diameter $\leq 10\mu\text{m}$)
SO ₂	Sulfur Dioxide
NO ₂	Nitrogen Dioxide
O ₃	Ozone
CO	Carbon Monoxide

II. LITERATURE REVIEW

Air pollution is an important environmental and public health issue, with extensive evidence in multiple studies of its negative impacts. The *World Health Organization* (WHO) estimates that air pollution is responsible for 6.7 million premature deaths every year, with a considerable portion happening in South Asia [1]. In India, air quality has become highly critical, and cities like Delhi, Mumbai, and Kolkata suffer from hazardous levels of air often due to road transport emissions, industrial processes, and construction materials dust [2]. 2024 progress report on the *National Clean Air Programme* (NCAP) indicates that out of 227 cities with over 75% data availability in 2023, 78 NCAP cities and 118 non-NCAP cities exceeded the NAAQS for PM₁₀ levels. [3].

While much research has been devoted to urban air pollution, rural areas are relatively under-studied, even though increasing evidence shows that pollution spillover, agriculture, and local emissions all play important roles in affecting air quality in these areas. Research indicates that rural sources of air pollution—like biomass burning, dust storms, and pesticide application—can result in PM_{2.5} and ozone concentrations equal to those found in urban areas [4]. A study conducted by Sujan Khanal et al. highlighted trans-boundary pollution and crop residue burning as major factors contributing to severe seasonal pollution episodes, particularly in North India [5].

There have been various studies of urban air pollution employing varied methodologies. AQI sensitivity in Tehran was investigated by Fotouhi et al., who found that pollutant-specific statistical parameters affect reported air quality, and maximum values tend to overestimate the severity of pollution [6]. Yadav and Toshniwal carried out a comparative study of the air quality in four large Indian cities, applying Principal Component Analysis (PCA) and time series models to discern seasonal patterns of major pollutants [7]. Likewise, Nandanwar and Chauhan examined improvements in air quality during India’s COVID-19 lockdown and recorded a substantial decline in PM_{2.5} and PM₁₀ concentrations due to diminished vehicle and industrial activity [8].

As a result of wide-ranging urban research, very little work exists on rural air pollution. Borghi et al. stressed the necessity of monitoring rural-specific pollutants and their emissions, mainly from plowing, harvesting, and the application of pesticides, which emit particulate matter and ammonia [4]. Mota-Bertran et al. also investigated the impact of air pollution on mental health among children in rural Catalonia, correlating

pollutant exposure with behavioral disorders, anxiety, and developmental difficulties [9]. Yet, most rural studies are based on sparse monitoring networks, constraining their capability to observe fine-scale spatio-temporal patterns.

Urban-rural pollution spillover is still an unexplored topic. Mohajeri et al. estimated indoor air pollution exposures, which identified large urban-rural differences, with rural households having greater PM_{2.5} exposure from biomass burning and the use of traditional fuels [10]. Likewise, a *Greenpeace India* report identified that 89% of seven major Indian cities’ monitoring stations had been surpassing WHO air pollution standards, highlighting vehicular emissions as a prevailing NO₂ source [11]. These studies do not include comparative urban-rural studies examining pollution gradients in terms of geographical proximities.

Advanced modeling approaches have been utilized in recent studies to analyze air pollution. Multivariate clustering has been used by Rahman and Khatun to classify Asian cities by the intensity of pollution and to identify gross differences between East and South Asia [12]. Jagtap et al. analyzed deep learning models for predicting air quality, using CNN and RNN and contrasting them with standard statistical methods [13]. Few of these methods have been used specifically in urban-rural comparisons of air quality, and there is a lack of understanding about pollution trends between these environments with the passage of time.

Improved rural monitoring has also been emphasized in some reports. Sharma et al. traced air quality trends in 336 Indian cities from 1987 to 2019 and showed that whereas levels of SO₂ reduced, PM₁₀ concentrations went up, most notably in the Indo-Gangetic Plain [14]. In addition, a *Down to Earth* report revealed that rural PM_{2.5} concentrations in India averaged 46.4 $\mu\text{g}/\text{m}^3$ in 2022—almost on par with urban averages—contradicting the notion that rural environments have cleaner air [15]. However, rural environments are still underrepresented in national monitoring networks and are constrained by limited data availability and the formulation of targeted mitigation measures.

Although earlier studies have widely examined air pollution in urban and, to a lesser degree, rural areas, there is a significant paucity of holistic spatio-temporal analyses comparing both environments. The literature hardly measures urban-rural pollution differences or analyzes spillover impacts with strong statistical and machine learning methods. Bridging these gaps is important for developing *region-specific air quality management policies* that take into account both urban emissions and rural sources of pollution.

III. PROPOSED ARCHITECTURE

The air quality assessment framework (Figure 1) is a multilevel system comprising data collection, pre-processing, calculation of AQI, and statistical analysis in order to analyze urban-rural pollution differences methodically.

Real-time pollutant concentration data (PM_{2.5}, PM₁₀, CO, NO₂, SO₂, O₃) are retrieved from the Open-Meteo API, with

redundancy checks ensuring reliability. Pre-processing incorporates unit standardization, timestamp alignment, and hourly aggregation to daily AQI. The AQI is calculated based on Central Pollution Control Board (CPCB) standards, where sub-indices for pollutants are calculated and aggregated for comparison purposes.

The analytical framework utilizes spatio-temporal analysis to determine seasonality trends and measure urban-rural differences by the Pollution Disparity Index (PDI). Random Forest regression identifies major pollutants driving AQI variations, whereas K-Means clustering classifies areas according to pollution trends, allowing the detection of high-risk areas and the main pollutants present in those areas. This facilitates the establishment of regional-specific policy.

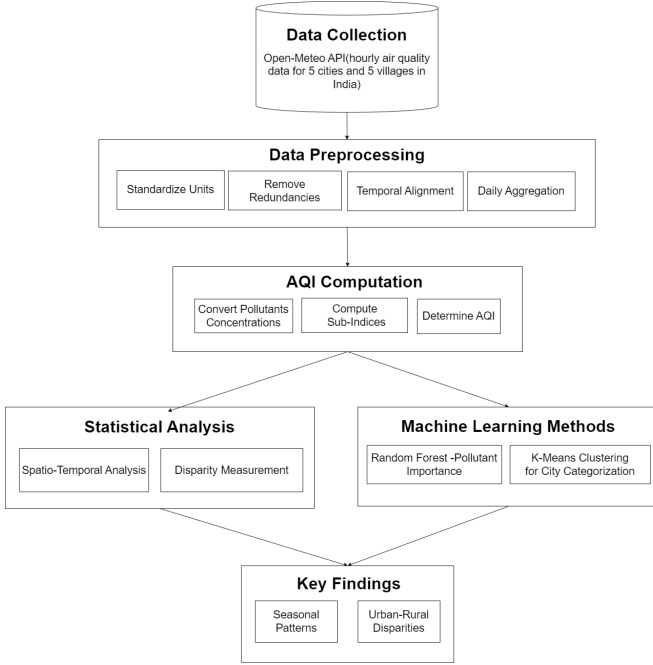


Fig. 1: Architecture Diagram

IV. METHODOLOGY

This study aims to analyze spatio-temporal variations in air quality across urban and rural locations in India using machine learning and statistical techniques. The key research questions guiding this work are:

- 1) How do pollutant concentrations differ between urban and rural environments across time and space?
- 2) What are the dominant pollutants influencing air quality trends in different regions?
- 3) How can machine learning and clustering methods improve the classification and prediction of air pollution patterns?
- 4) What is the extent of urban-rural pollution disparities, and in which seasons do they occur?

A. Data Collection and Preprocessing

To investigate urban-rural air quality disparities, we examined five city-village pairs in India: Chennai-Koonimedu

(Tamil Nadu), Delhi-Asawar (Delhi), Ahmedabad-Saputara (Gujarat), Kolkata-Bhalukhop (West Bengal), and Shillong-Mawlynnong (Meghalaya) (Table II). These city-village pairs were chosen to represent geospatial variability, involving coastal, interior, industrial, and hilly regions, for an equable comparison. Each rural site is a village with fewer than 10,000 residents, providing an instructive comparison of scale, infrastructure, and source of pollution to its city equivalent.

TABLE II: Urban and Rural Locations Across States/UTs in India Considered for the Study

State/Union Territory	City	Village
Tamil Nadu	Chennai	Koonimedu
Delhi & Uttar Pradesh	Delhi	Asawar
Gujarat	Ahmedabad	Saputara
West Bengal	Kolkata	Bhalukhop
Meghalaya	Shillong	Mawlynnong

The dataset spans September 1, 2022 – January 31, 2025, sourced from the Open-Meteo API, which provides hourly pollutant concentrations and meteorological data using location-specific geospatial interpolation. The pollutants analyzed include PM_{2.5}, PM₁₀, CO, NO₂, SO₂, and O₃, which are key indicators of air quality degradation.

Preprocessing Steps:

- **Unit Standardization:** CO concentrations were converted from µg/m³ to mg/m³ per CPCB guidelines.
- **Outlier Detection and Removal:** The Interquartile Range (IQR) and Z-score methods were used to detect and remove extreme sensor anomalies. IQR is effective for skewed data, while Z-score is suited for normal distributions, ensuring robust outlier handling.
- **Missing Data Handling:** No missing value imputation was required as the dataset was complete.

Following preprocessing, AQI was computed for each hourly record, resulting in 212,160 hourly AQI values. These were further aggregated into 8,841 daily AQI values per location, ensuring spatio-temporal consistency for comparative analysis.

B. Air Quality Index (AQI) Computation

AQI was calculated based on the sub-index approach as recommended by the Central Pollution Control Board (CPCB) of India. The approach converts raw concentrations of pollutants into separate sub-indices, which are subsequently employed to calculate the final AQI for a day. The calculation is based on piecewise linear interpolation between pre-defined breakpoint values.

The sub-index for each pollutant is computed as:

$$I_p = I_{low} + \frac{(I_{high} - I_{low})(C_p - C_{low})}{(C_{high} - C_{low})} \quad (1)$$

where:

- I_p = Sub-index of the pollutant
- C_p = Observed pollutant concentration

- C_{low}, C_{high} = Breakpoint concentration limits
- I_{low}, I_{high} = Corresponding AQI values at breakpoint limits

The final AQI for each location is determined by selecting the maximum sub-index across all pollutants:

$$AQI = \max(I_{PM_{2.5}}, I_{PM_{10}}, I_{NO_2}, I_{SO_2}, I_{CO}, I_{O_3}) \quad (2)$$

C. Spatio-Temporal Analysis of Air Quality Trends

To investigate long-term air quality variations, seasonal decomposition techniques were used to separate pollutant trends into:

- 1) Long-term progression (trends over months/years)
- 2) Seasonal fluctuations (recurring annual or monthly cycles)
- 3) Residual anomalies (unexpected deviations)

Moreover, air quality data were also combined at yearly, monthly, and weekday resolutions to capture intra-week and seasonal variations. Line plots and heatmaps were used for visualizing temporal dynamics of pollutants, indicating peak pollution events and trends.

D. Comparative Spatial Analysis of Pollutant Distributions in Urban and Rural Areas

A spatial analysis was used to evaluate differences in pollutant concentrations between urban and rural environments. To better examine spatial patterns, comparative analysis was carried out to analyze pollutant distributions in urban and rural settings. Urban settings were analyzed for transportation, industrial process, and construction emissions, while rural settings were analyzed for pollutant impacts of agricultural and natural processes. Further, correlation analysis was carried out to examine relationships between pollutants across different settings. The association between $PM_{2.5}$ and PM_{10} was examined to identify common emission sources, while O_3 - NO_2 interactions were analyzed to understand their role in photochemical reactions. These analyses provided insights into pollution dispersion and the necessity for spatially adaptive air quality management strategies.

E. Temporal Feature Importance Analysis

Random Forest regression models were used to evaluate the relative contributions of different pollutants to AQI variations over time. Monthly data subsets were used for training the models, ensuring sufficient data points for reliable feature importance estimation. Feature importance trends were assessed across different months and seasonal periods to provide insight into the changing significance of pollutants over time.

F. Clustering-Based Classification of Air Quality Patterns

K-Means clustering was employed to categorize locations based on their average AQI values from September 2022 to January 2025. The clustering results highlighted high-risk areas with persistently elevated AQI levels, indicative of severe pollution exposure. By segmenting locations into distinct clusters, this analysis facilitates a comparative assessment of pollution levels, supporting targeted air quality management and policy formulation.

G. Pollution Disparity Index (PDI) for Urban-Rural Comparisons

To quantify urban-rural pollution gradients, a Pollution Disparity Index (PDI) was developed, defined as:

$$PDI = \left(\frac{AQI_{urban} - AQI_{rural}}{AQI_{rural}} \right) \times 100\% \quad (3)$$

where:

- AQI_{urban} = Daily AQI in urban locations.
- AQI_{rural} = Daily AQI in rural locations.

Higher PDI values indicate greater urban-rural pollution gaps, necessitating region-specific regulatory actions.

V. RESULTS AND KEY FINDINGS

A. Spatio-Temporal Analysis

TABLE III: Location-wise AQI Values

Location	AQI
Ahmedabad	71.89
Asawar	147.23
Bhalukhop	90.45
Kolkata	139.66
Chennai	82.12
Delhi	153.00
Koonimedu	84.98
Mawlynnong	82.91
Saputara	87.04
Shillong	82.91

The Air Quality Index (AQI) data varies across locations, with Delhi (153.00), Asawar (147.23), and Calcutta (139.66) showing the highest pollution levels, posing health risks, especially for sensitive groups. In contrast, Ahmedabad (71.89), Chennai (82.12), Koonimedu (84.98), Mawlynnong (82.91), Saputara (87.04), and Shillong (82.91) have relatively lower AQI, indicating moderate air quality (Table III)

Urban areas, such as Delhi and Calcutta, have higher pollution due to vehicular emissions, industrial activities, and seasonal factors [16]. Meanwhile, cleaner air in regions like Shillong and Mawlynnong can be linked to lower industrialization and dense green cover [17].

The presence of coastal winds in Chennai and Koonimedu aids in pollutant dispersion, preventing excessive accumulation [18]. Additionally, Delhi's air quality worsens during winter due to temperature inversions that trap pollutants near the surface [19].

B. PDI-Disparity Measurements

Table IV shows that Ahmedabad has significantly lower pollution compared to Delhi (-53.01%) and Asawar (-51.17%), reflecting differences in industrial activity, traffic emissions, and environmental policies.

TABLE IV: Pairwise Location PDI

Location 1	Location 2	PDI (%)
Ahmedabad	Asawar	-51.17
Ahmedabad	Bhalukhop	-20.52
Ahmedabad	Kolkata	-48.53
Ahmedabad	Chennai	-12.46
Ahmedabad	Delhi	-53.01
Ahmedabad	Koonimedu	-15.41
Ahmedabad	Mawlynnong	-13.29
Ahmedabad	Saputara	-17.40
Ahmedabad	Shillong	-13.29
Asawar	Bhalukhop	62.77
Asawar	Kolkata	5.42
Asawar	Chennai	79.29
Asawar	Delhi	-3.77
Asawar	Koonimedu	73.25
Asawar	Mawlynnong	77.58
Asawar	Saputara	69.16
Asawar	Shillong	77.58
Bhalukhop	Kolkata	-35.23
Bhalukhop	Chennai	10.15
Bhalukhop	Delhi	-40.88
Bhalukhop	Koonimedu	6.44
Bhalukhop	Mawlynnong	9.10
Bhalukhop	Saputara	3.93
Bhalukhop	Shillong	9.10
Kolkata	Chennai	70.07
Kolkata	Delhi	-8.72
Kolkata	Koonimedu	64.34
Kolkata	Mawlynnong	68.45
Kolkata	Saputara	60.46
Kolkata	Shillong	68.45
Chennai	Delhi	-46.33
Chennai	Koonimedu	-3.37
Chennai	Mawlynnong	-0.95
Chennai	Saputara	-5.65
Chennai	Shillong	-0.95
Delhi	Koonimedu	80.04
Delhi	Mawlynnong	84.54
Delhi	Saputara	75.79
Delhi	Shillong	84.54
Koonimedu	Mawlynnong	2.50
Koonimedu	Saputara	-2.36
Koonimedu	Shillong	2.50
Mawlynnong	Saputara	-4.74
Mawlynnong	Shillong	0.00
Saputara	Shillong	4.98

Delhi exhibits the highest pollution levels, particularly in contrast to Mawlynnong (84.54%) and Koonimedu (80.04%), reinforcing its reputation for severe air pollution.

Mawlynnong and Shillong show identical pollution levels (PDI = 0.00%), suggesting similar environmental conditions. Their low pollution is attributed to dense vegetation, minimal industrial activity, and high rainfall aiding pollutant dispersion.

Chennai has a mixed pollution profile, with relatively small differences compared to Koonimedu (-3.37%), Mawlynnong (-0.95%), and Shillong (-0.95%).

Asawar exhibits drastically higher pollution than Bhalukhop (62.77%) and other locations, making it one of the most polluted areas, second only to Delhi. Factors such as industrial emissions, inadequate waste management, and pollution spillover contribute to its high AQI levels.

C. Pollutants Importance Across Rural and Urban Areas

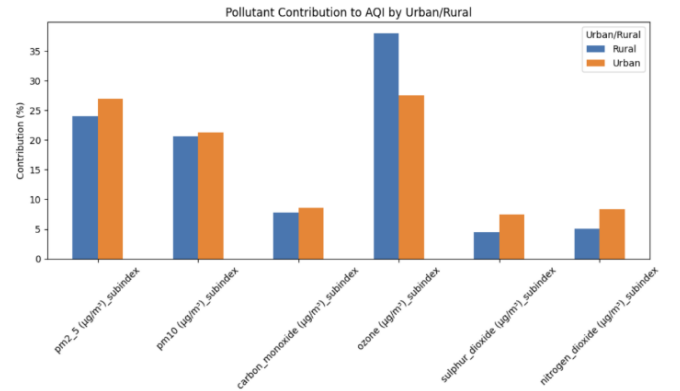


Fig. 2: Pollution Contribution to AQI for Urban VS Rural

Figure 2 highlights the differences in pollutant contributions to AQI between urban and rural areas. Ozone (O₃) is highest in rural regions, while PM_{2.5} and PM₁₀ are significant in both. Other pollutants, such as CO, SO₂, and NO₂, have lower contributions. Urban areas generally exhibit higher pollution levels, except for ozone.

Ozone is elevated in rural areas due to reduced nitric oxide (NO) emissions, which typically break down ozone in cities. Winds transport ozone from urban sources, allowing it to accumulate in rural areas. Strong sunlight accelerates ozone formation, and fewer airborne pollutants mean less interference. Additionally, vegetation releases biogenic VOCs that react with transported NO_x, further increasing ozone [20], [21]. Unlike the protective ozone layer in the stratosphere, ground-level ozone is a pollutant affecting air quality. It can impact plant growth, reduce crop yields, and contribute to environmental stress in forests and ecosystems [20].

PM_{2.5} and PM₁₀ are crucial in air quality studies due to their widespread presence and environmental effects. PM_{2.5} consists of fine particles that remain suspended in the air for long periods, affecting visibility and atmospheric composition. PM₁₀, being larger, contributes to dust and air pollution. Urban areas see higher PM_{2.5} and PM₁₀ levels from traffic, industry, and construction, while rural sources include agriculture, biomass burning, and dust storms. Monitoring these particles

provides essential data for understanding pollution trends and developing air quality management strategies.

D. K-Means Based Clustering of Rural and Urban Areas

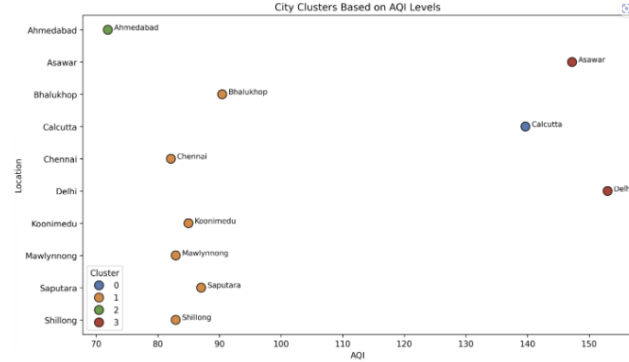


Fig. 3: K-Means categorized clusters

Figure 3 shows the spatial grouping of cities by their Air Quality Index (AQI) values with the K-Means algorithm. The chart classifies the cities into four clusters, each represented by a different color, to facilitate a comparative examination of air quality rates.

Cluster 0, which includes Calcutta, is the moderately high AQI level areas. Cluster 1, with cities like Bhalukhop, Chennai, Koonimedu, Mawlynnong, Saputara, and Shillong, reflects comparatively lower pollution areas. Cluster 2, represented by Ahmedabad, is areas with the lowest AQI, reflecting comparatively good air quality. Cluster 3, with very polluted cities like Delhi and Asawar, reflects critical air quality conditions.

E. Monthly AQI Trends

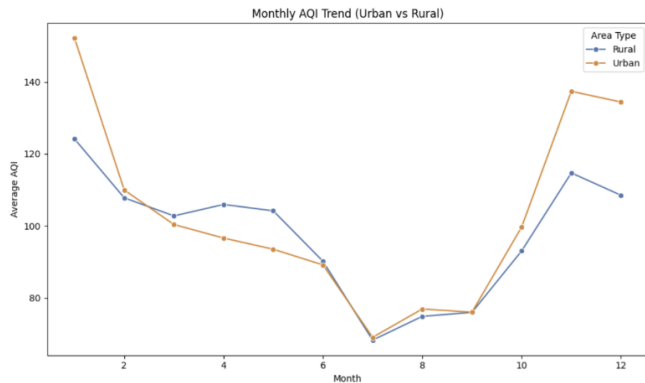


Fig. 4: Monthly AQI Trend (Urban Vs Rural)

Figure 4 presents the monthly Air Quality Index (AQI) trends for urban and rural areas, revealing significant seasonal variations in air pollution levels. Throughout the year, urban AQI remains consistently higher than rural AQI, underscoring the impact of anthropogenic emissions in densely populated regions. Both regions exhibit a declining trend in AQI from

January to July, followed by a sharp increase from October to December.

The mid-year dip, most pronounced during July, coincides with the monsoon period, when enhanced rainfall serves as a natural scavenger, eliminating atmospheric pollutants via wet deposition. Strong surface winds and atmospheric turbulence also enhance pollutant dispersion, resulting in better air quality [22]. On the other hand, the steep increase in AQI during December and January can be explained by winter meteorological conditions, such as temperature inversions that accumulate pollutants at the surface [23]. Further, rising emissions due to biomass burning, domestic fuel, transportation activity, and industrial activities aggravate pollution levels. Agricultural residue burning and the depositing of particulate matter in rural areas also witness higher AQI in winter months [24].

F. Monthly Pollutant Variations Across Rural and Urban Areas

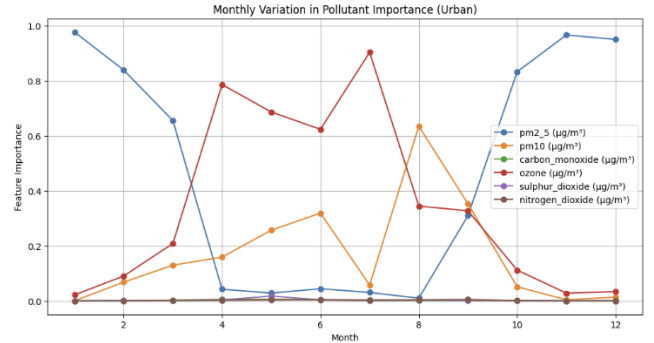


Fig. 5: Monthly Pollutant Variations - Urban

Figure 5 illustrates the monthly variation in pollutant importance in urban areas, highlighting how different pollutants influence air quality throughout the year. PM_{2.5} has the highest impact during winter months, likely due to increased emissions from vehicles, industries, and lower atmospheric dispersion, while ozone (O₃) peaks in summer, driven by higher temperatures and photochemical reactions [20]. PM₁₀ fluctuates, with a notable rise in late summer, possibly due to construction activities and dust. Other pollutants like carbon monoxide (CO), sulfur dioxide (SO₂), and nitrogen dioxide (NO₂) remain relatively low in importance.

Compared to rural areas, urban regions show a stronger influence of PM₁₀, emphasizing the need for targeted pollution control strategies to address seasonal variations effectively.

Figure 6 illustrates the monthly variation in pollutant importance for rural areas, emphasizing the influence of different pollutants on air quality throughout the year. Notably, ozone (O₃) and PM_{2.5} exhibit the most significant variations, with ozone concentrations peaking from March to July, while PM_{2.5} levels dominate in January and towards the end of the year.

Other air pollutants, such as PM₁₀, carbon monoxide (CO), sulfur dioxide (SO₂), and nitrogen dioxide (NO₂), have relatively lower significance with minor changes over time. The predominance of ozone in warmer months indicates a high

influence of photochemical reactions induced by sunlight and temperature [20], [21], while the increase in PM_{2.5} in colder months can be explained by enhanced biomass burning, decreased atmospheric dispersion, and stagnant weather patterns.

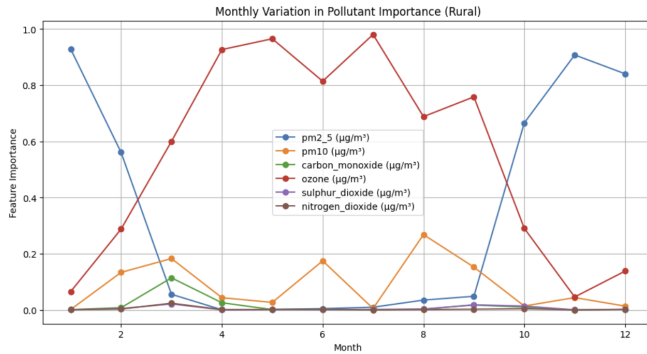


Fig. 6: Monthly Pollutant Variations - Rural

VI. CONCLUSION AND FUTURE WORKS

The current research presents an overall analysis of air quality trends in urban and rural areas, reflecting the changing character of pollution patterns through time. Based on a thorough spatiotemporal analysis, it illustrates evident seasonal and spatial differences, pointing out the dynamic interaction among environmental conditions and emission sources.

Results indicate that the highest concentrations of PM_{2.5} in urban regions occur in winter, predominantly due to the effects of transport, industrial processes, and atmospheric conditions. Conversely, ozone variations over rural areas during summer indicate that there is intense reliance on agricultural activities and photochemical processes. Pollution, nonetheless, is not limited to the point of pollution. Urban effluents penetrate beyond urban limits, reaching out to surrounding villages and small-scale settlements. This highlights the interactive nature of air quality issues such that variations in one area would have quantifiable effects on others. To solve these problems needs a greater scope that goes beyond urban-oriented interventions to cover regional approaches.

Identifying particular clusters of pollution and measures of urban-rural differences in the study underlines the demand for inclusive methods in air quality management. Dissimilarity between pollutions between regions implies that corrective measures ought to be tailored at the region-based level without dependency on any panacea measure to be equally adopted everywhere. Expanding strategies against pollution into a wider than only urban domain helps in improving the air sustainably and spatially equitably.

In addition to increasing air quality evaluations, the combination of satellite-based air quality data with ground observations can increase spatial coverage and accuracy. Moreover, machine learning models can also be used to forecast future AQI trends on the basis of meteorological conditions and emission sources to facilitate proactive air quality management.

Extending the research to a wider set of regions and including real-time monitoring systems would give greater insights

into pollution dynamics and control measures. In addition, analyzing the effect of policy interventions, including vehicular restrictions and industrial regulations, can provide useful insights for policymakers in formulating effective pollution mitigation strategies.

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